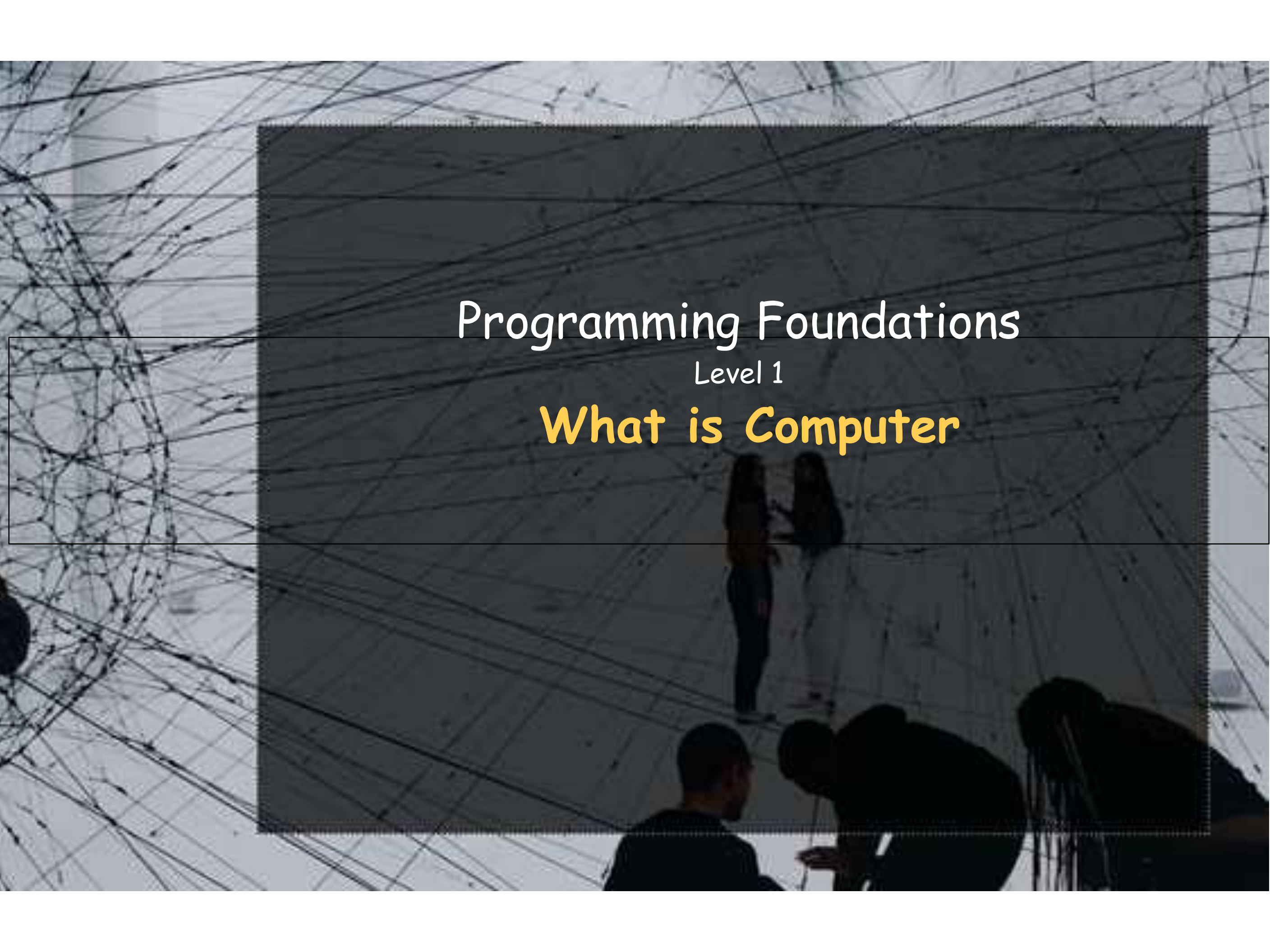




Programming Foundations Level 1

The background of the slide features a complex network of thin, dark lines crisscrossing a light gray background, resembling a neural network or a web. In the lower portion, there are dark silhouettes of people: two individuals standing and talking in the center, and a group of three people in the foreground, one of whom appears to be looking at a screen or device.

Programming Foundations

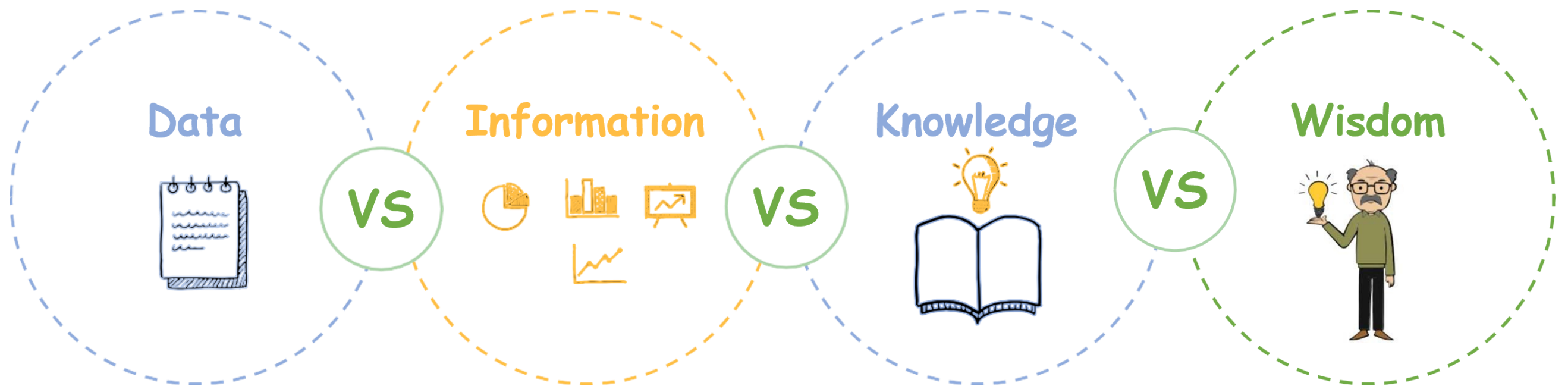
Level 1

What is Computer

Before we know what is computer?

Let us remember...

What is the difference between...

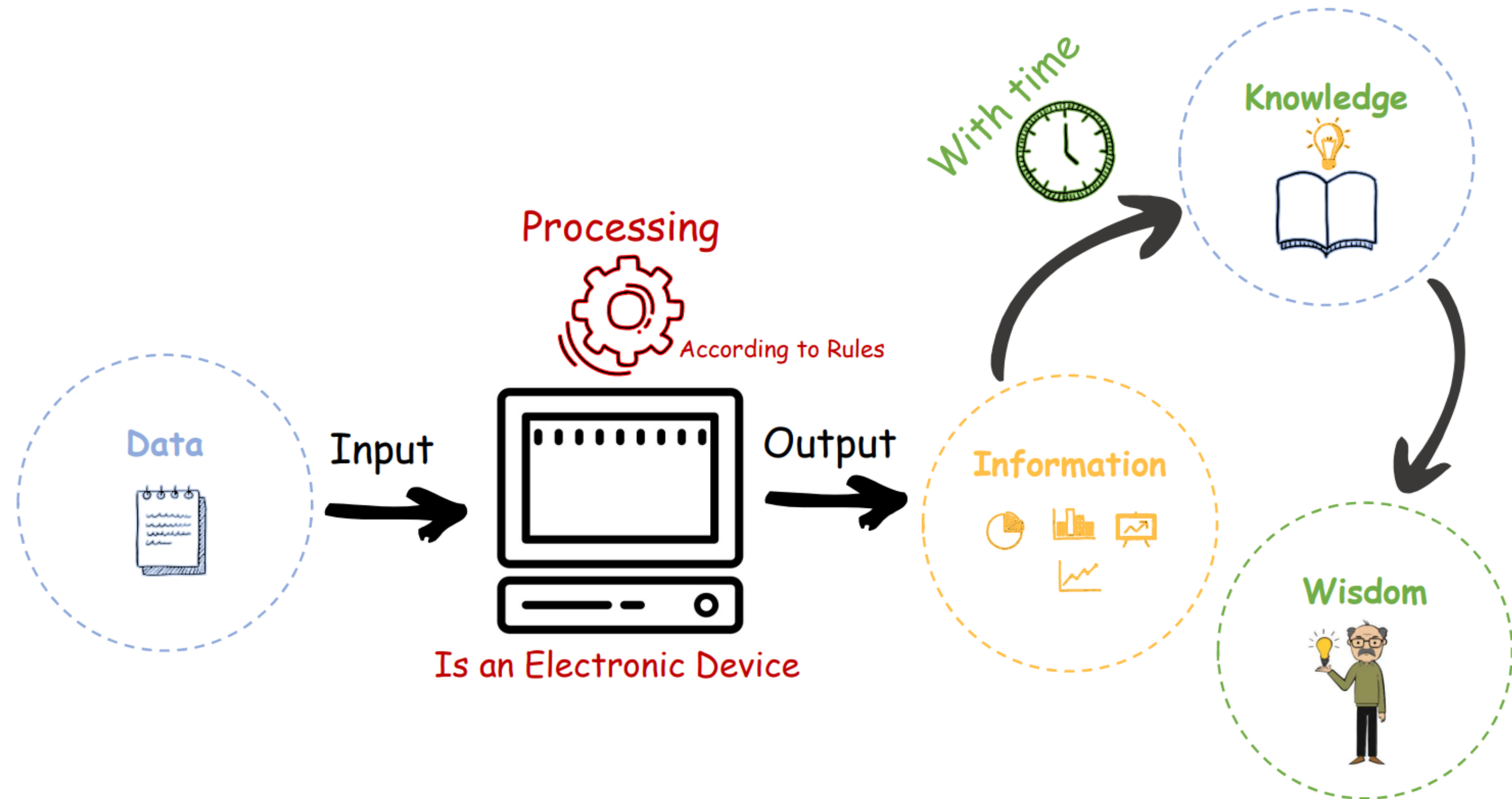


Computer is:

A **computer** is more than a box on a desk—it's an **electronic device** that accepts **data (input)**, **processes** it under **control of instructions**, produces **information (output)**, and **stores** that information for future use.

We'll unpack what that means, then tour the **core units** (Input, Output, Memory, CPU), explain **software vs hardware**, compare **CPU vs GPU**, and demystify **32-bit vs 64-bit**.

What is computer?



Computer is:

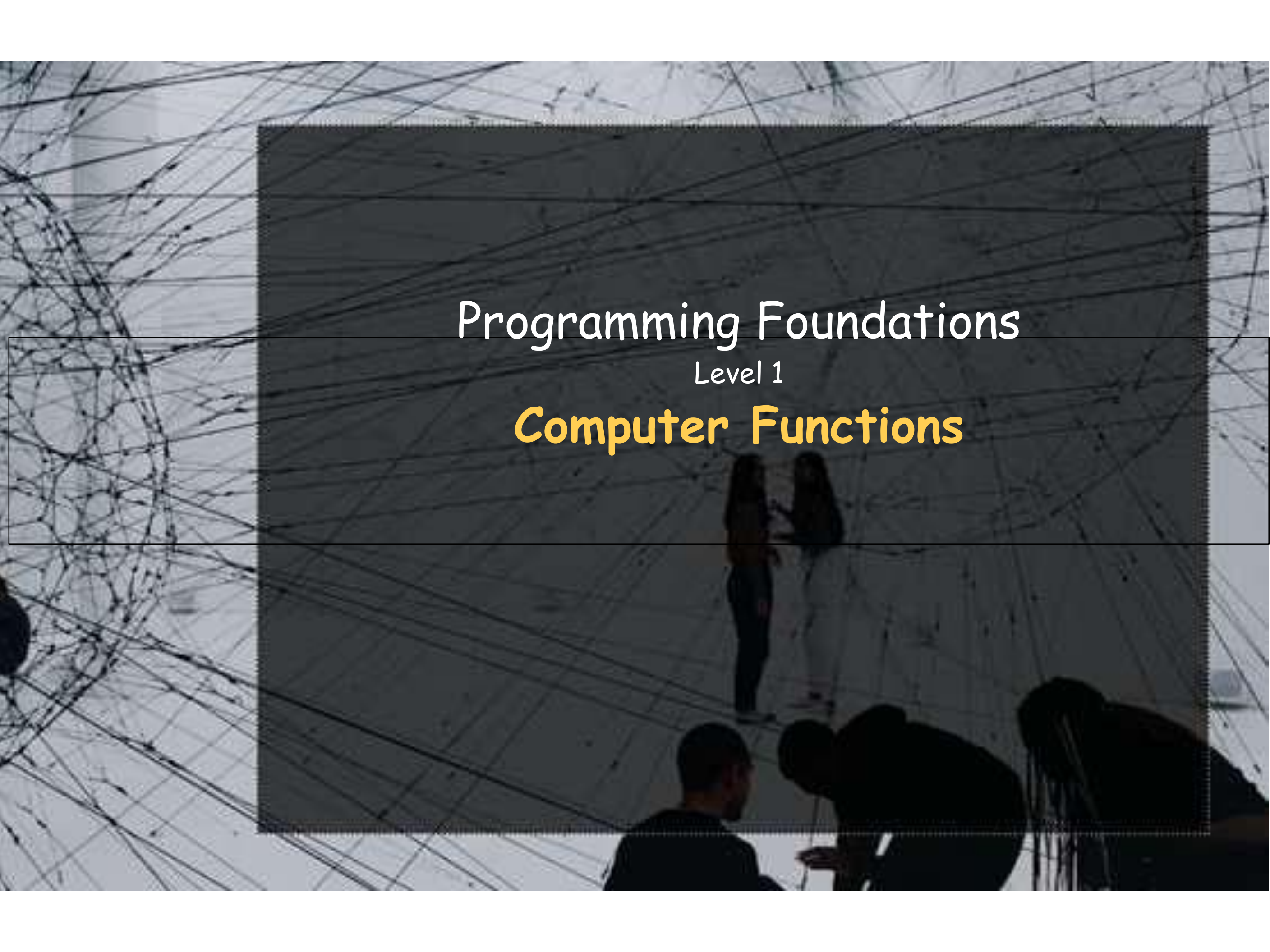
A computer is:

an Electronic Device 

operating under the control of instructions (programs) 

that accepts data (input), processes data according to rules, produces information (output), and stores information for future use .

Processing Model: Input → Processing → Output (Data → Information → Knowledge → Wisdom perspective).

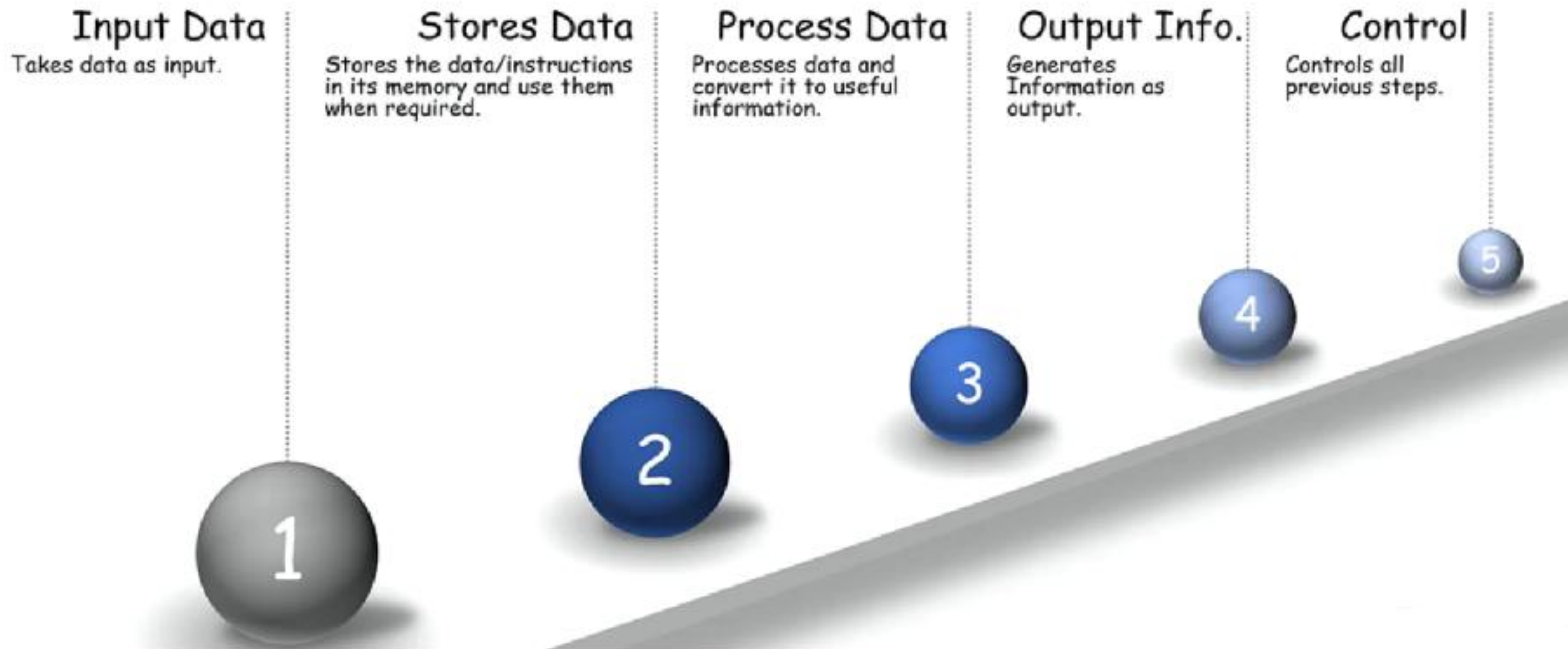
The background of the slide is a dark, abstract image. It features a dense, intricate network of thin, light-colored lines that resemble a complex web or a neural network. In the lower portion of the image, there are dark silhouettes of several people. Two people are standing in the center, facing each other, while others are visible in the foreground, some appearing to be looking at a screen or device. The overall aesthetic is technical and modern.

Programming Foundations

Level 1

Computer Functions

Functionalities of Computer

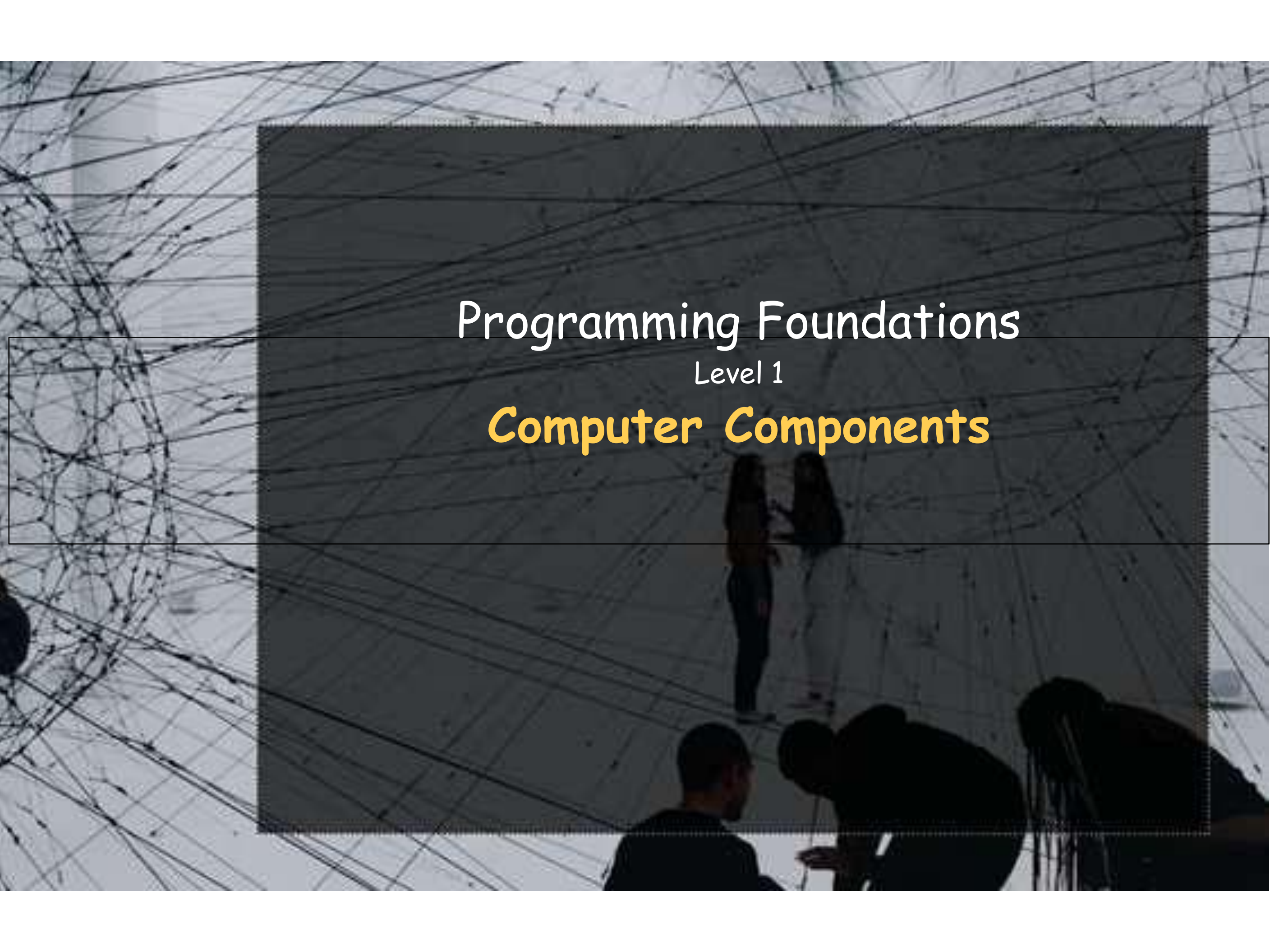


Computer is:

- 1 Input data - Takes data in.
- 2 Stores data - Keeps data/instructions in memory for later.
- 3 Process data - Converts data into useful information.
- 4 Output information - Presents results.
- 5 Control - Coordinates and controls all previous steps.

Computer is:

- Electronic Device.
- Operating under the control of instructions.
- Accepts Data (Input).
- Processes Data according to certain rules.
- Produces Information (Output).
- Stores information for future use.



Programming Foundations

Level 1

Computer Components

Functionalities of Computer



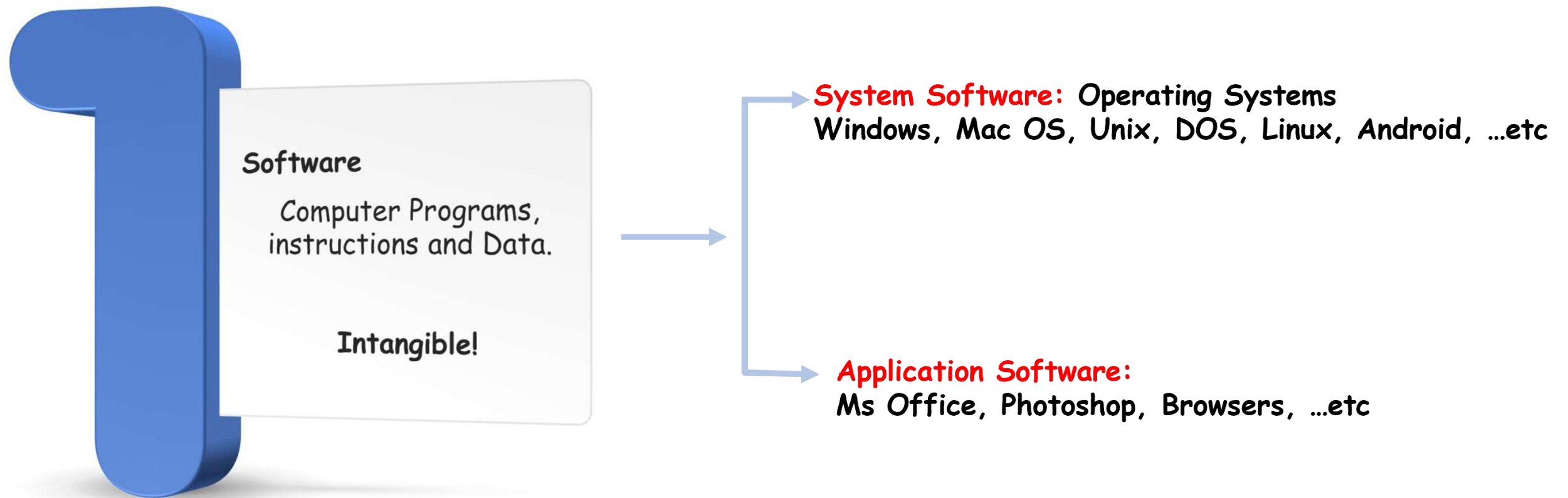
The background of the slide features a complex, abstract pattern of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a meeting. A large, dark, semi-transparent rectangular box is centered on the slide, serving as a backdrop for the text.

Programming Foundations

Level 1

Computer Components Software

Software





Programming Foundations

Level 1

Computer Components Hardware

Hardware

2

Hardware

Collection of physical elements (tangible objects)

Tangible!



Computer Case



Monitor



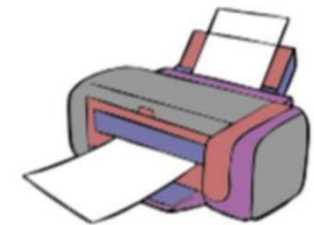
Speakers



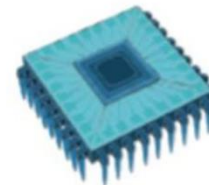
Keyboard



Mouse



Printer



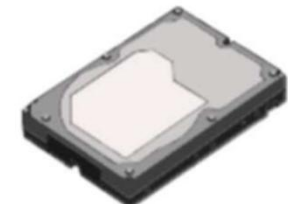
Processor (CPU)



Motherboard



RAM



Hard Drive



Programming Foundations

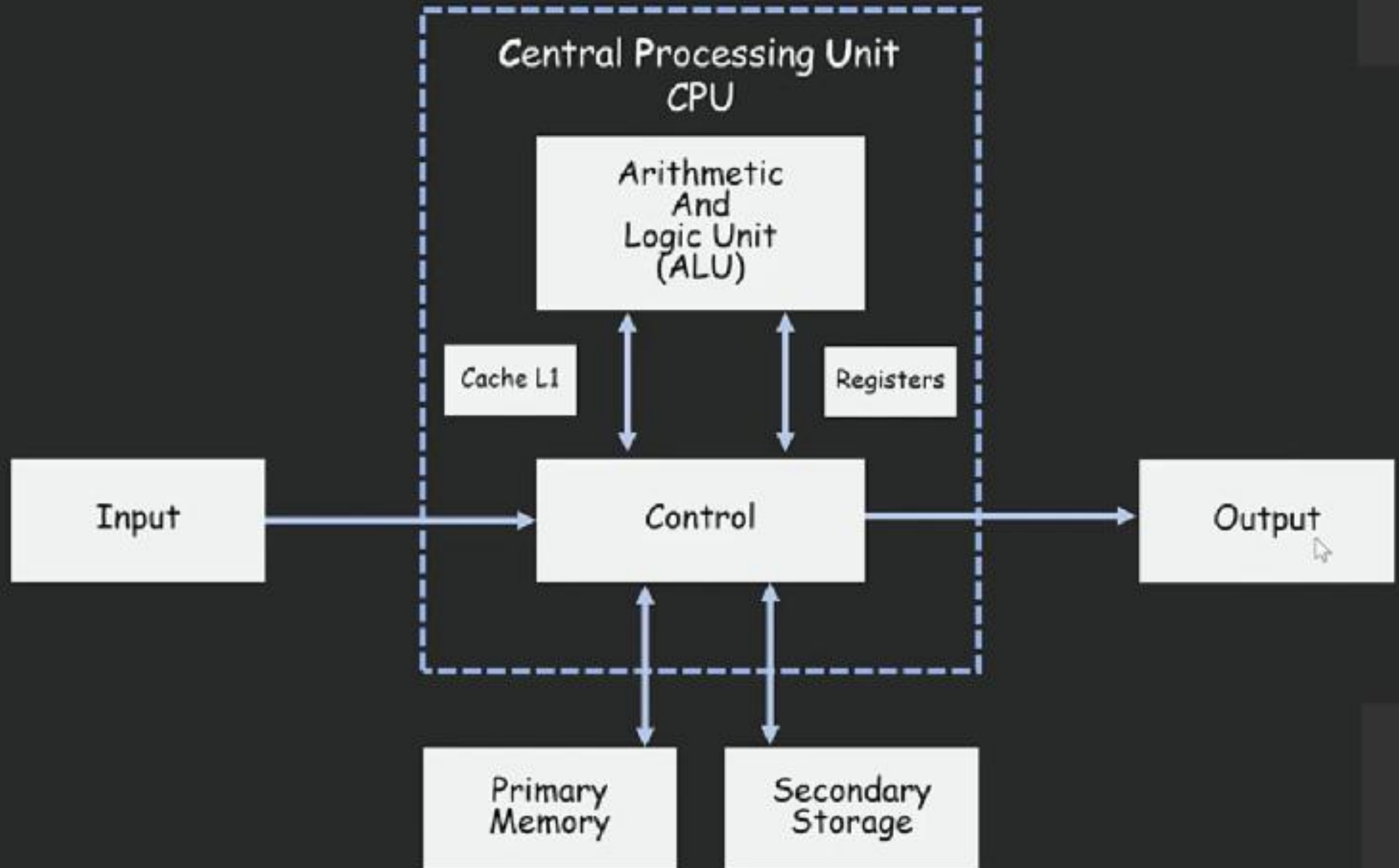
Level 1

Computer Units

Computer Units



Computer Units



Computer Units:

- Input
- Output
- Primary Memory (RAM/ROM)
- Secondary Storage (long-term)
- Central Processing Unit (CPU) (ALU, Control, Cache & Registers)

A more detailed view shows **CPU** at the center with **ALU, Control Unit, Cache/Registers**, connected to **Input/Output** and **Primary/Secondary Memory**.

Computer Units Input



Input: any device that provides data or signals to the computer.

Input

1

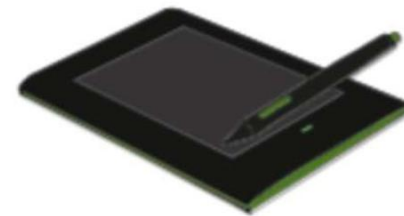
Any device that provides data or signal to computer.



Keyboard



Mouse



Touch Screen
& Pens

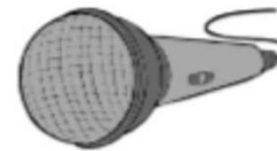


Joystick

Camera



Remote



Mic



Scanner

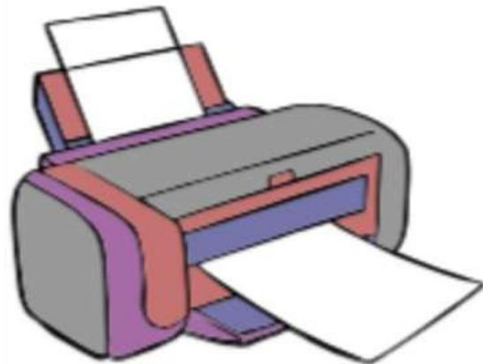
Computer Units Output



Output: any device used to **communicate results** (monitor, printer, speakers).



**Any device used
to communicate
results.**



Printer



Monitor

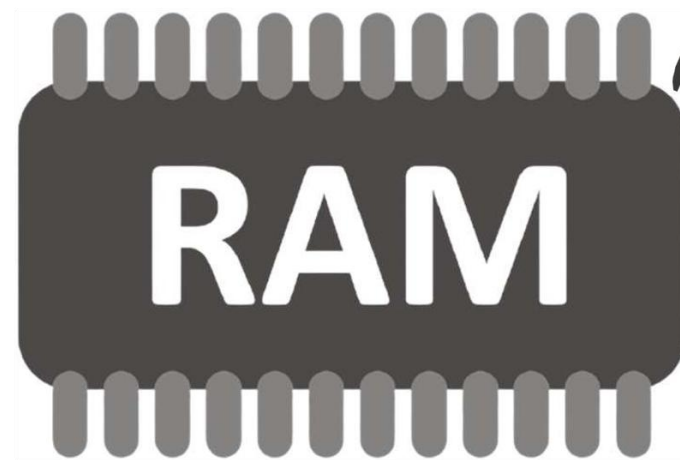


Speaker

Computer Units Primary Memory

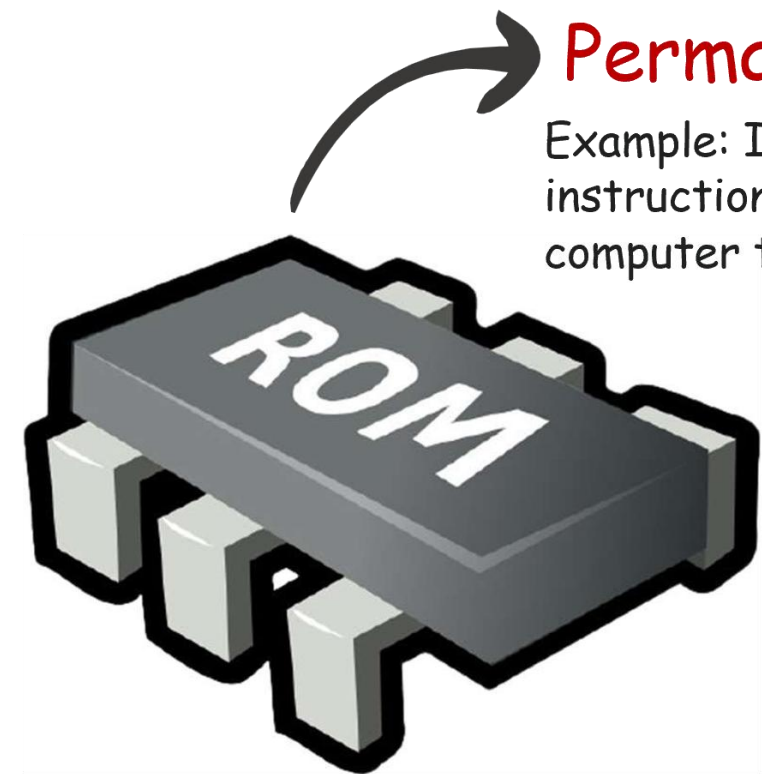


RAM & ROM



Random Access Memory

Temporary



Permanent

Example: It stores the instructions for the computer to start up.

Read Only Memory

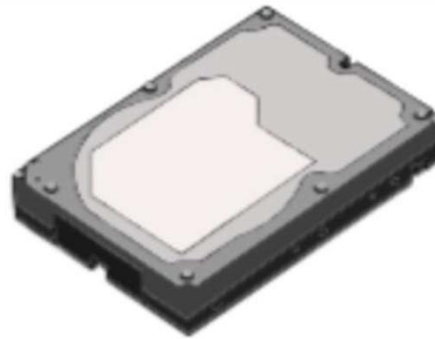
- RAM (Random Access Memory): temporary working memory.
- ROM (Read-Only Memory): permanent—stores startup/boot instructions

Computer Units Secondary Storage

Secondary Storage

4

Stores data
permanently



Hard Drive



CD



Memory Card



Flash Memory

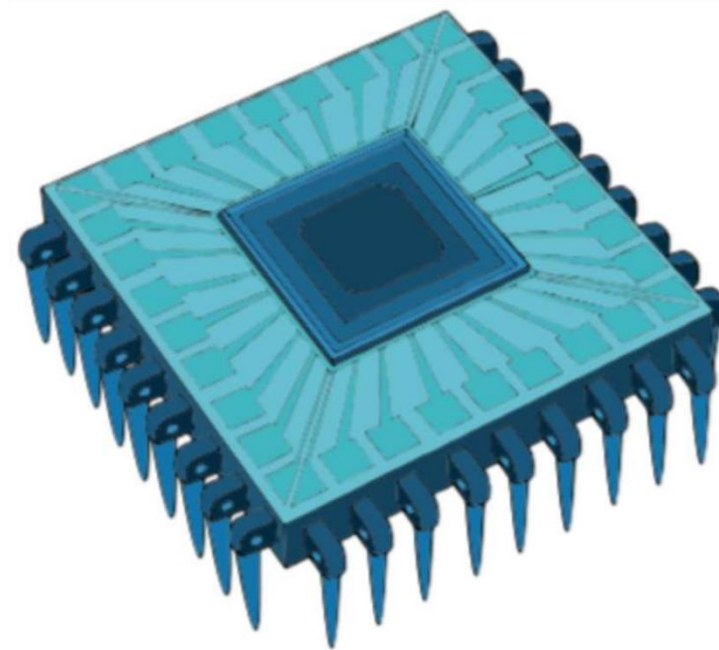
■ Long-term storage; stores data permanently (e.g., SSD/HDD).

Computer Units CPU

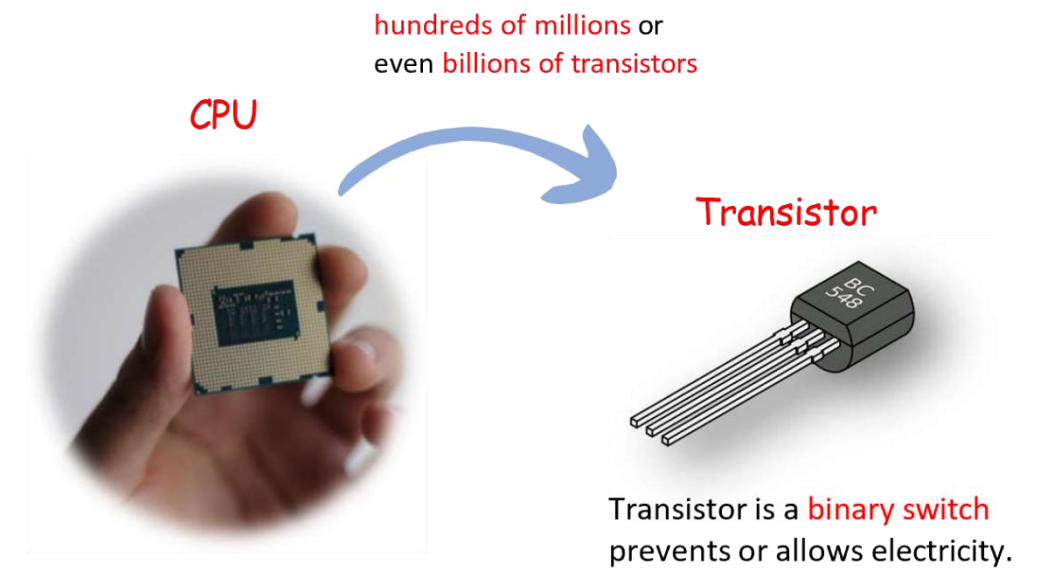


It is responsible for all functions and processes.

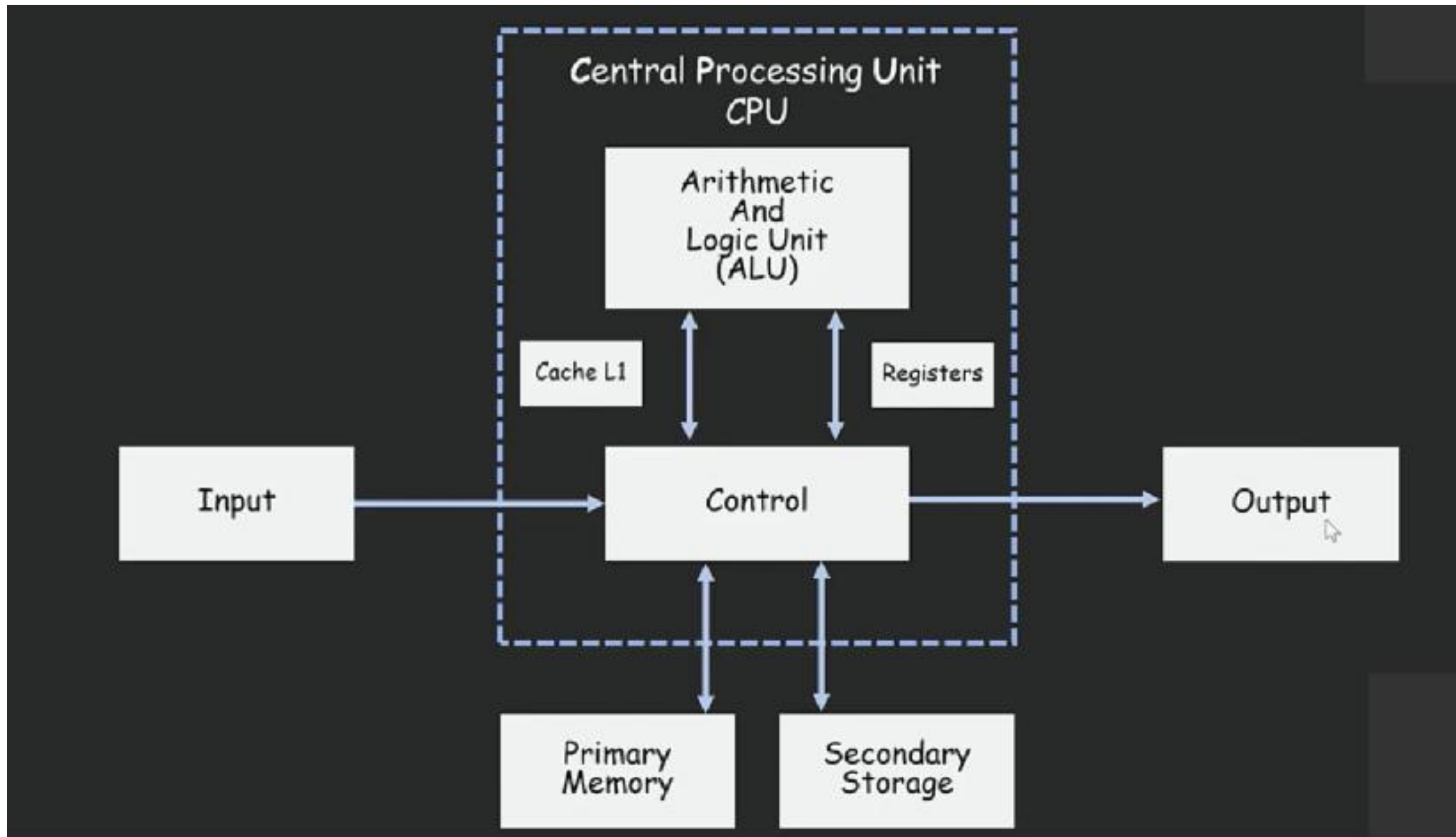
Known as microprocessor or processor



Central Process Unit (CPU)



Computer Units



- Also called **microprocessor/processor**; responsible for **all functions and processes** at the instruction level.
- ALU (Arithmetic Logic Unit)**: arithmetic (add, sub, mul, div) + logical operations (compare numbers/letters/characters).

Arithmetic and logical Unit - ALU:

- Executes **Arithmetic** and **logical** operations.
 - Arithmetic **calculations** (Addition, Subtraction, Multiplication, Division). Logical
 - Operations (**Compare** numbers, letters, characters).
- **Control Unit (CU)**: controls & coordinates components, reads data from memory, sends to ALU/registers, instructs hardware to perform operations.

Control Unit - CU:

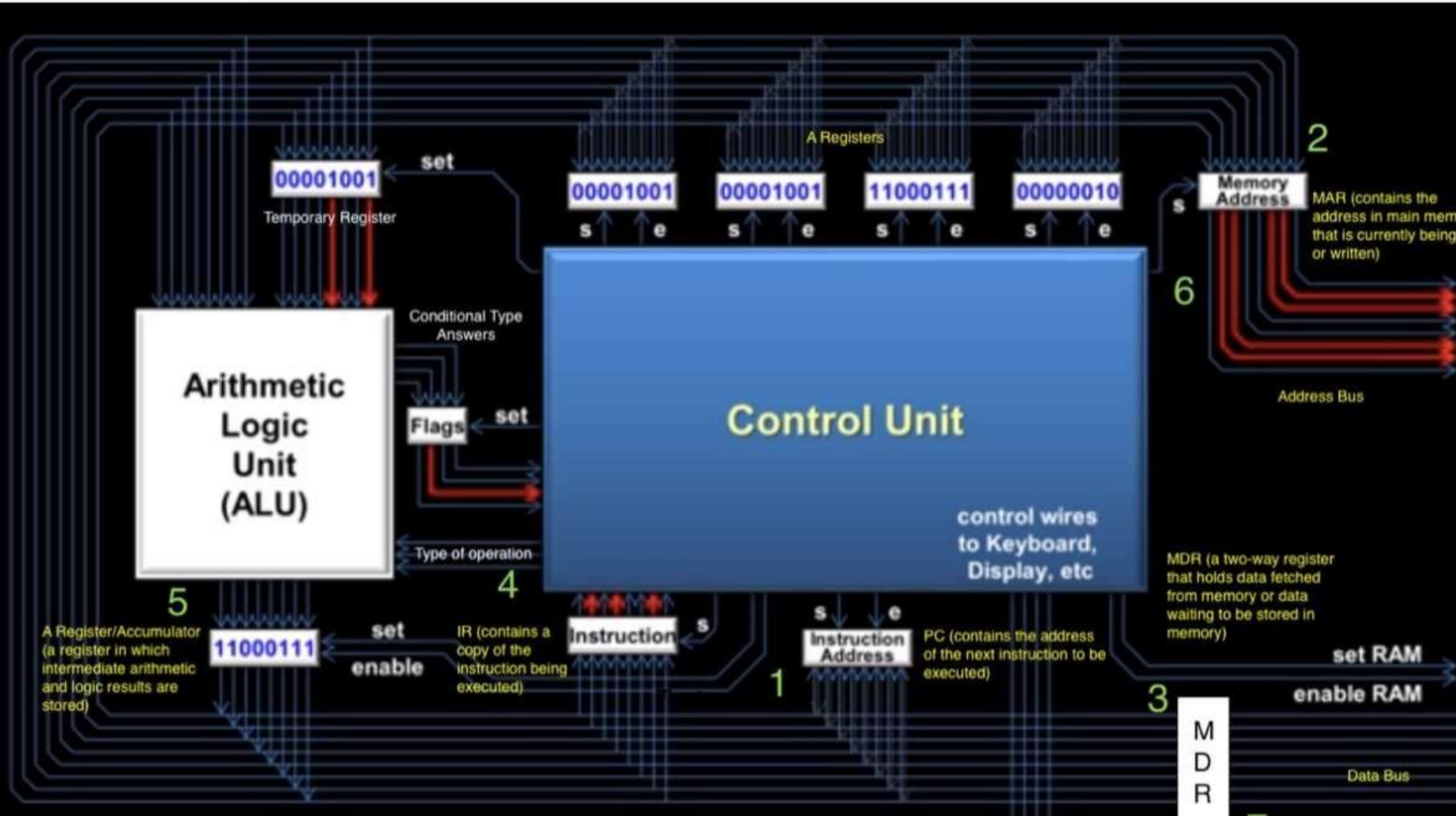
- Controls and **coordinates** computer components.
 - Reads data from memory
 - Sends data to ALU or register
 - Instructs hardware to perform the requested operation
- **Cache & Registers:** **high-speed** memory locations inside the CPU, directly accessible by the processor.

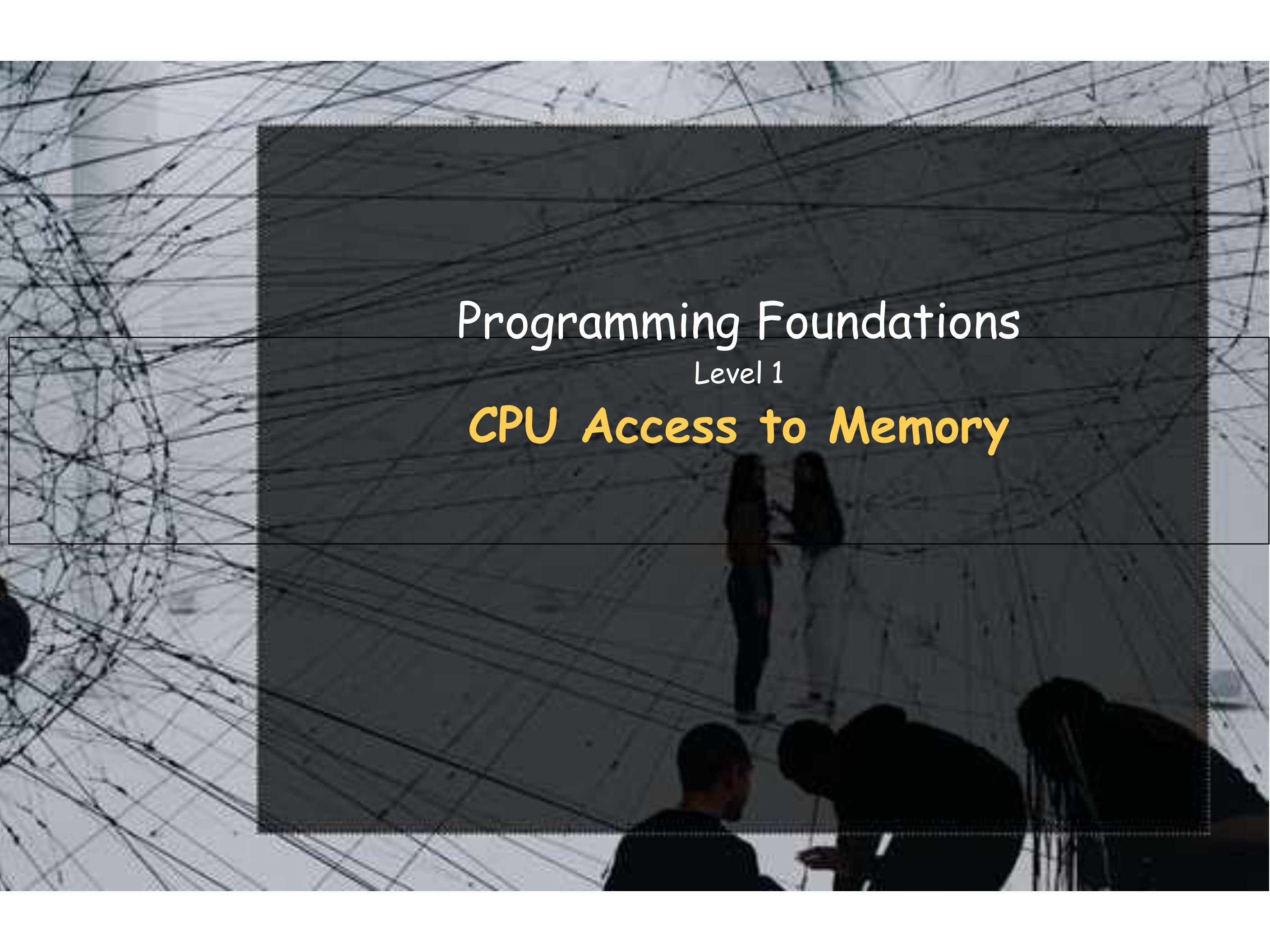
Cache & Registers:

- Cache is a **High-Speed memory** inside the CPU chip.
 - Registers are **Higher-Speed memory** inside the CPU chip.
- These are memory locations that can be **directly accessible by processor**.

CPU internal architecture:

at this level of your learning journey just take a look on this picture, don't dive into details.



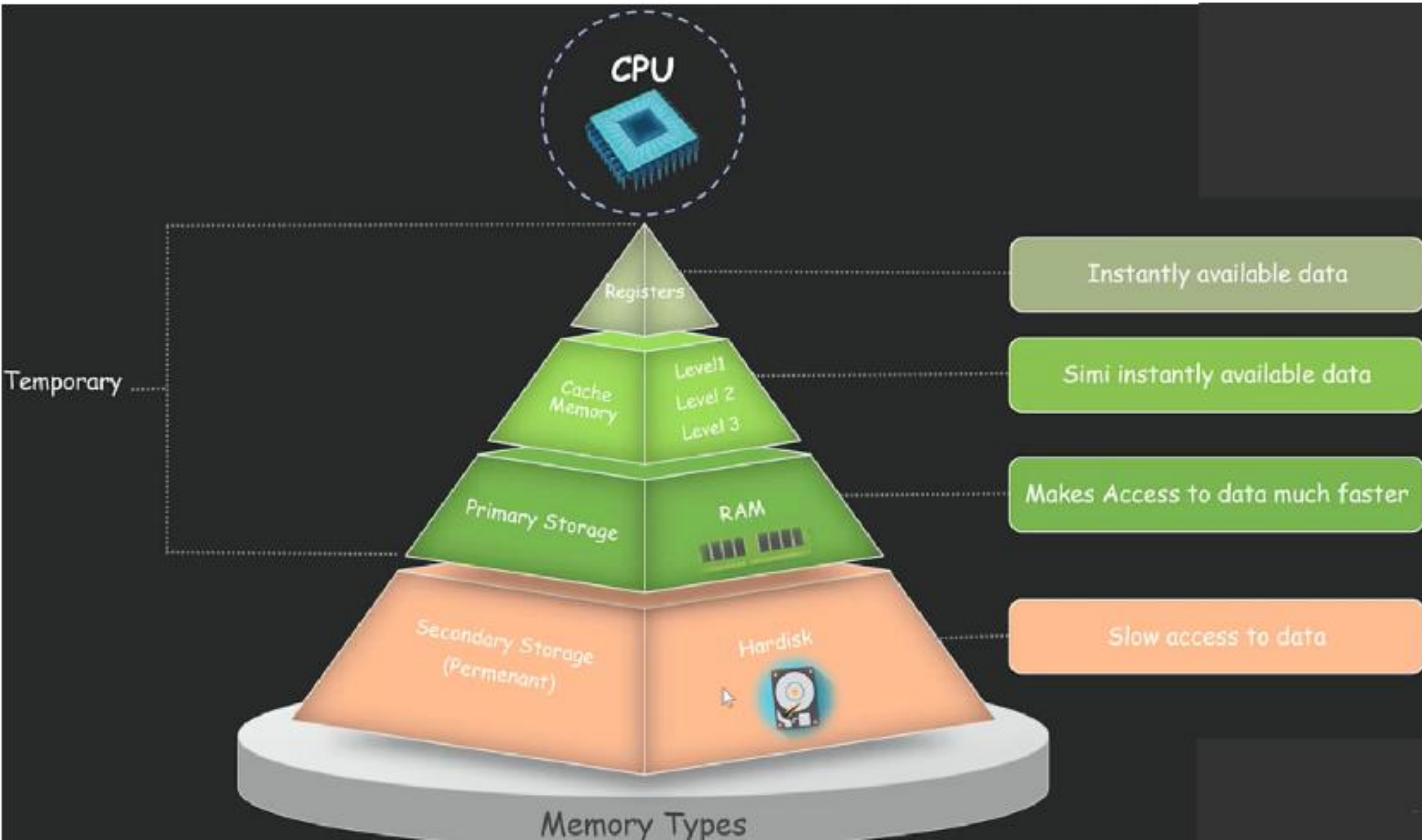
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Programming Foundations

Level 1

CPU Access to Memory

CPU Access to Memory: Types & Speed



CPU Access to Memory: Types & Speed

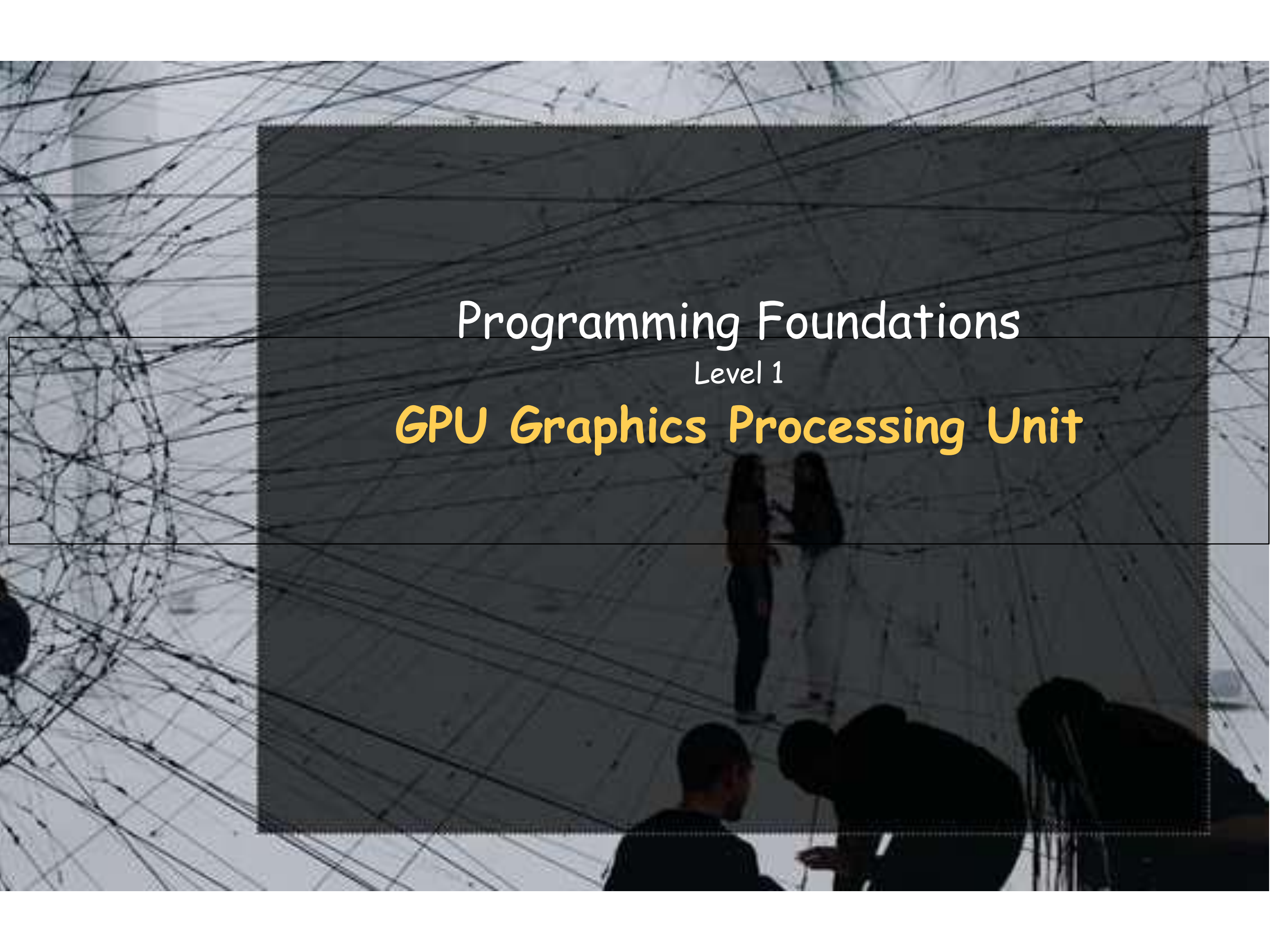
- **Registers** (inside CPU) → instant data

- **Cache (L1...)** → very fast

- **Main Memory (RAM)** → fast

- **Secondary Storage** → slower

“Instantly/simi-instantly” available data increases as you move closer to the CPU (Registers/Cache).

The background of the slide is a dark, abstract image. It features a dense, light-colored grid of lines that intersect to form a pattern of small squares, similar to a wireframe or a digital mesh. In the lower portion of the image, there are dark silhouettes of several people. Two people are standing in the center, facing each other, while others are visible in the foreground and background, some appearing to be in motion or interacting. The overall tone is technical and modern.

Programming Foundations

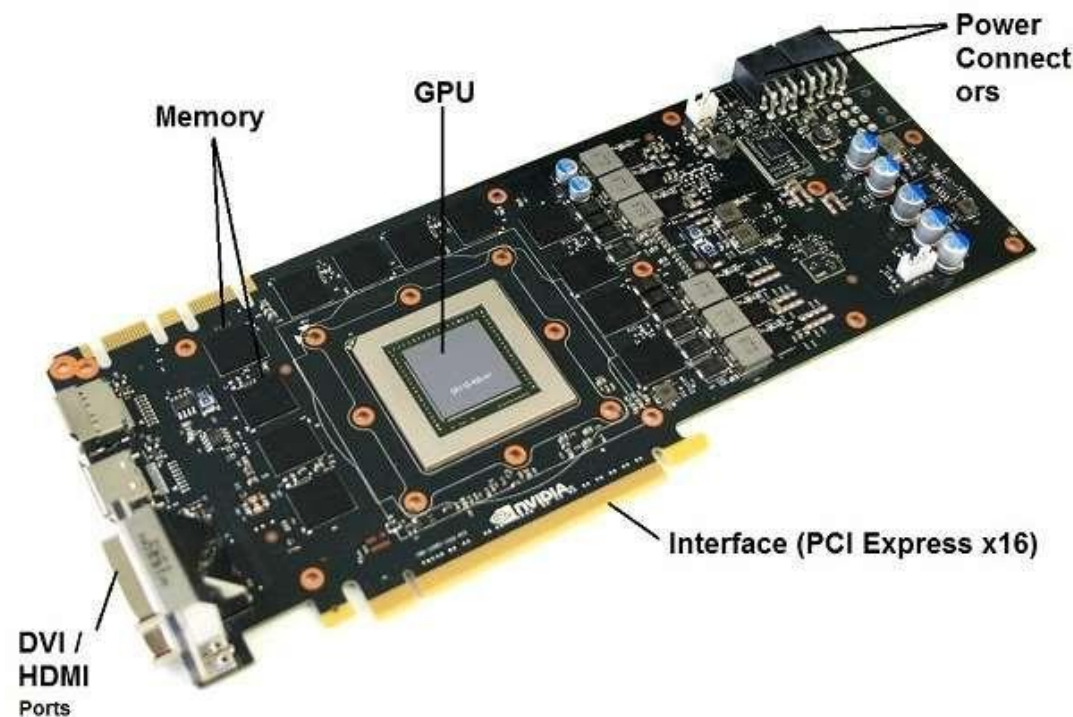
Level 1

GPU Graphics Processing Unit

What is GPU?

- GPU stands for graphics processing unit.
- You'll also see GPUs commonly referred to as graphics cards or video cards!
- But **it is not graphic card, it's only one part of it.**

GPU (Graphics Processing Unit): often called "graphics/video card," but the **GPU** is just one part of the card.



GPU vs CPU

- CPU is designed to handle tasks quickly but are limited in the concurrency of tasks that can be running.
- A GPU is designed to quickly render high-resolution images and video concurrently.
- **CPU:** designed to handle tasks quickly but with limited concurrency.
- **GPU:** designed to render high-resolution images/video concurrently (high parallelism).

The background of the slide is a dark, abstract image. It features a dense, intricate network of thin, light-colored lines that resemble a complex web or a neural network. In the lower portion of the image, there are dark silhouettes of several people. Two people are standing in the center, facing each other, while others are visible in the foreground, some appearing to be looking at a screen or device. The overall tone is technical and modern.

Programming Foundations

Level 1

How Computer Deals with data?

Computers Come in 32-bit and 64-bit Versions (x86 vs x64)

Modern computers are designed with **different architectures** that define how much data they can handle at once and how much memory they can address. The two most common are **32-bit** and **64-bit**.

💡 Address Space Capacity

📊 $2^{32} = 4,294,967,296 \text{ bytes} \approx 4 \text{ GB}$

👉 This is the maximum memory (RAM) a 32-bit system can address.

📊 $2^{64} = 18,446,744,073,709,551,616 \text{ bytes} \approx 16 \text{ Exabytes (EB)}$

👉 This shows the massive difference in capacity between 32-bit and 64-bit systems

🏷️ Naming Conventions

🟠 32-bit → often called **x86** or **x86-32**

🟠 64-bit → often called **x64** or **x86-64**

⚙️ Performance Difference

🟠 A 64-bit CPU can process twice as many bits per instruction cycle as a 32-bit CPU.

🟠 It can handle larger data types, larger files, and more RAM efficiently.



Note:

If you install a 32-bit Operating System on a 64-bit CPU, the system will still work — but you'll be limited to running 32-bit software only, and your system can't use all available memory.

RAM Limitations

-  32-bit OS: can address up to $2^{32} = 4,294,967,296$ bytes ≈ 4 GB
-  64-bit OS: can theoretically address 2^{64} bytes ≈ 16 Exabytes
-  If your computer has 4 GB RAM or more, you must use a 64-bit CPU and 64-bit OS to utilize it fully.

✨ Summary

- ✓ Computers are built on either **32-bit (x86)** or **64-bit (x64)** architecture.
- ✓ **64-bit systems** support more RAM, larger files, and faster performance.
- ✓ **Compatibility matters** — choose your OS and apps according to your CPU type

Activity

Instructions:

- 1 Draw a block diagram with **Input, Output, Primary Memory, Secondary Storage, CPU (ALU, CU, Cache/Registers)** and label each.
- 2 Given: "Load image from SSD → process filter → display on monitor." Map each step to **units** and indicate **where the data resides** at each step.
- 3 Your PC has **8 GB RAM**—explain **why** a **64-bit OS** is recommended and what happens if a 32-bit OS is installed.

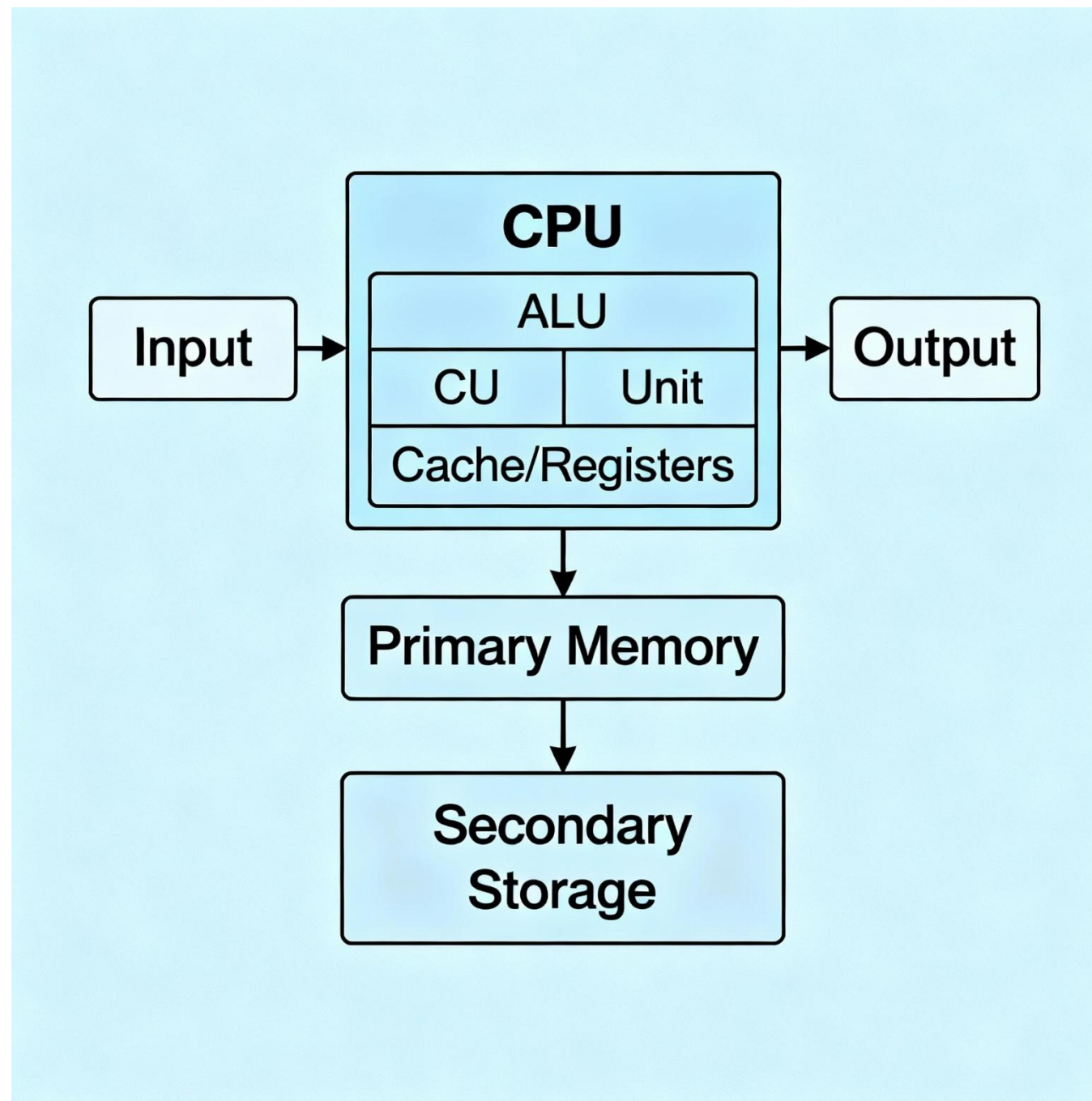


Activity Solution



Solution:

- 1 Draw a block diagram with Input, Output, Primary Memory, Secondary Storage, CPU (ALU, CU, Cache/Registers) and label each.

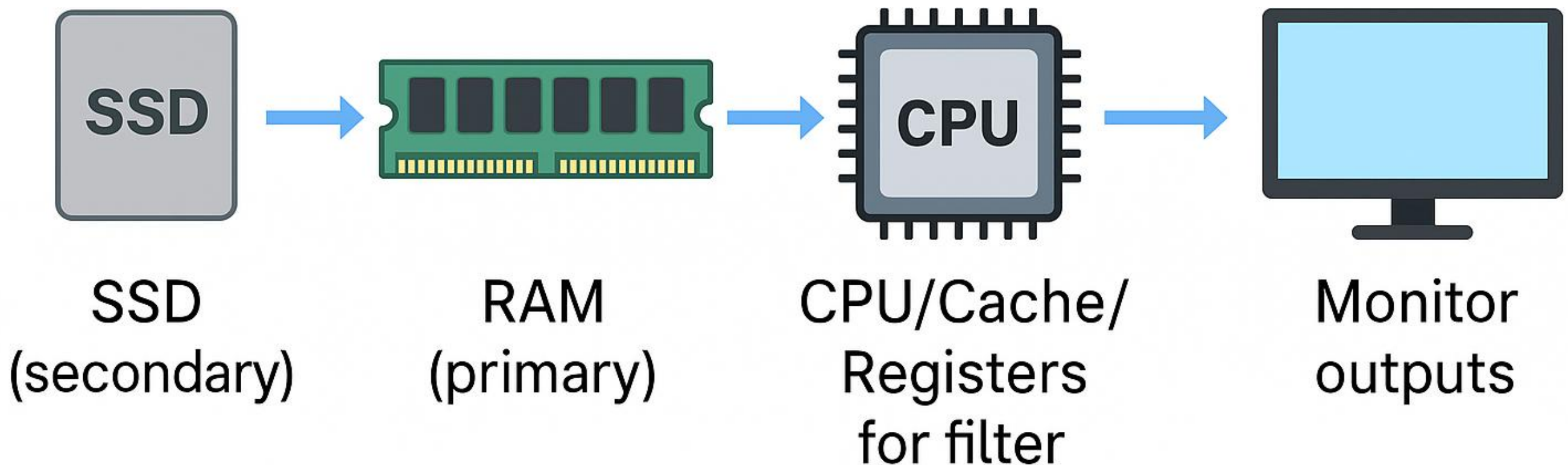


🤔 Activity Solution

📋 Solution:

2 Given: "Load image from SSD → process filter → display on monitor." Map each step to units and indicate where the data resides at each step.

🟡 **SSD** (secondary) → **RAM** (primary) → **CPU/Cache/Registers** for filter
→ **GPU/CPU** renders → **Monitor** outputs.





Activity Solution



Solution:

- 3** Your PC has 8 GB RAM—explain why a 64-bit OS is recommended and what happens if a 32-bit OS is installed.
- 64-bit OS:** can address >4 GB; 32-bit OS limited to ~4 GB, so it can't fully utilize 8 GB. A 64-bit CPU + 64-bit OS uses the memory effectively.

Conclusion

- ✓ A computer is an electronic device guided by programs to convert input data into useful output information and store it.
- ✓ Its units—Input, Output, Memory (RAM/ROM), Storage, CPU (ALU, CU, Cache/Registers)—work together under control.
- ✓ Software directs hardware; GPU complements CPU for parallel graphics.
- ✓ 64-bit architectures deliver larger address spaces and higher throughput vs 32-bit.

Thank you 😊